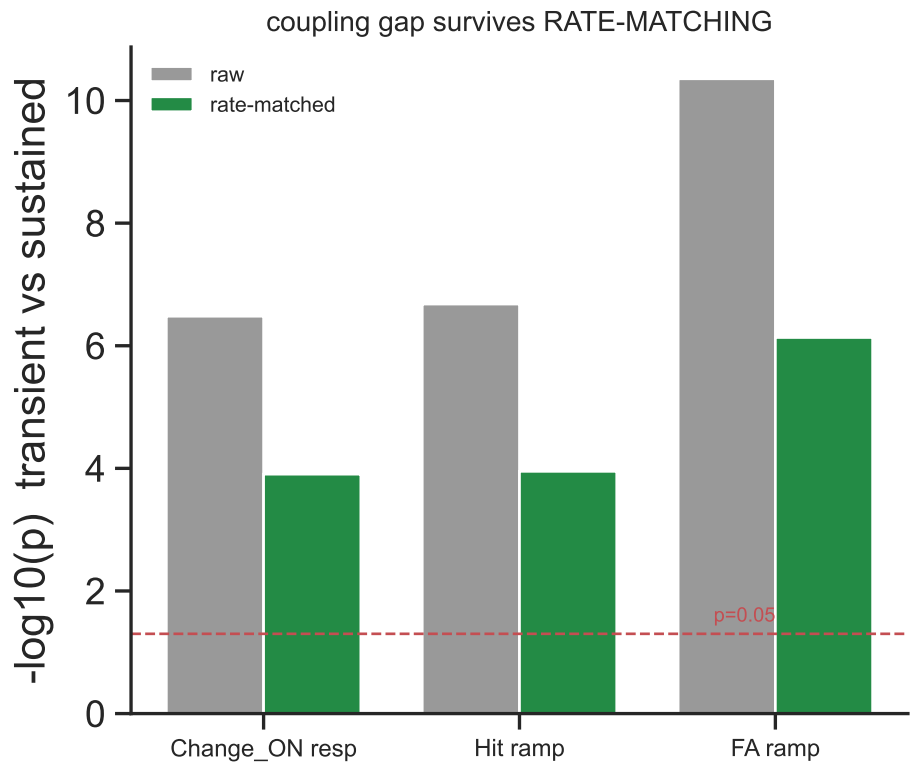
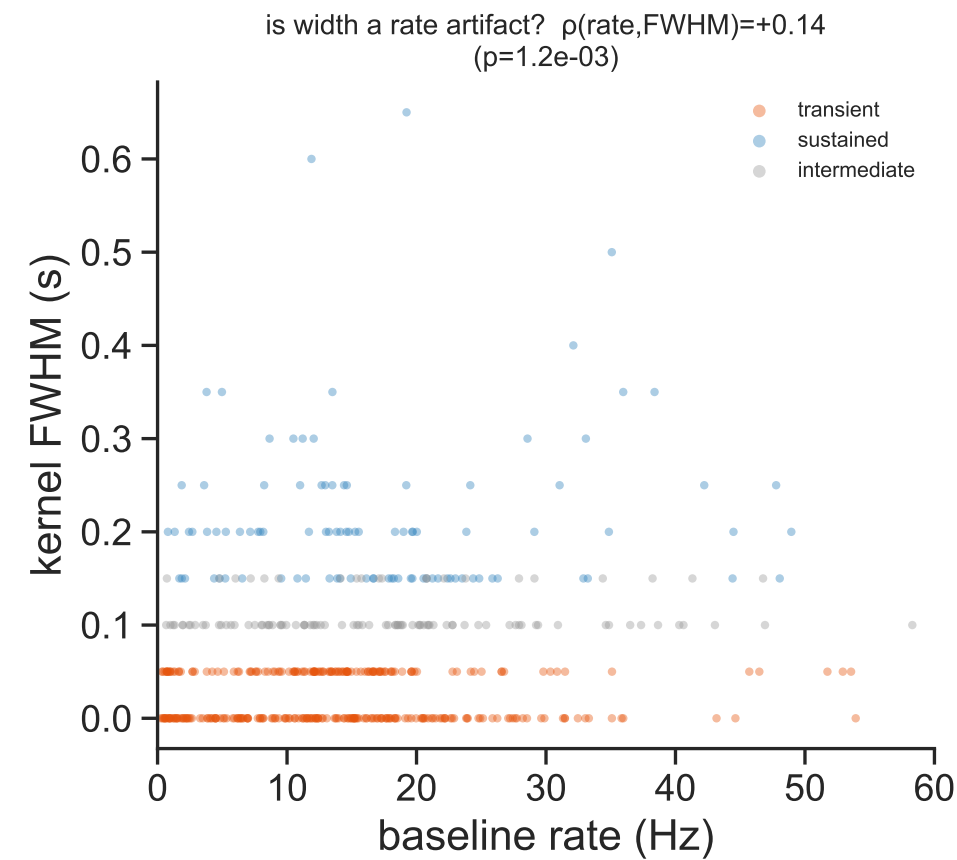
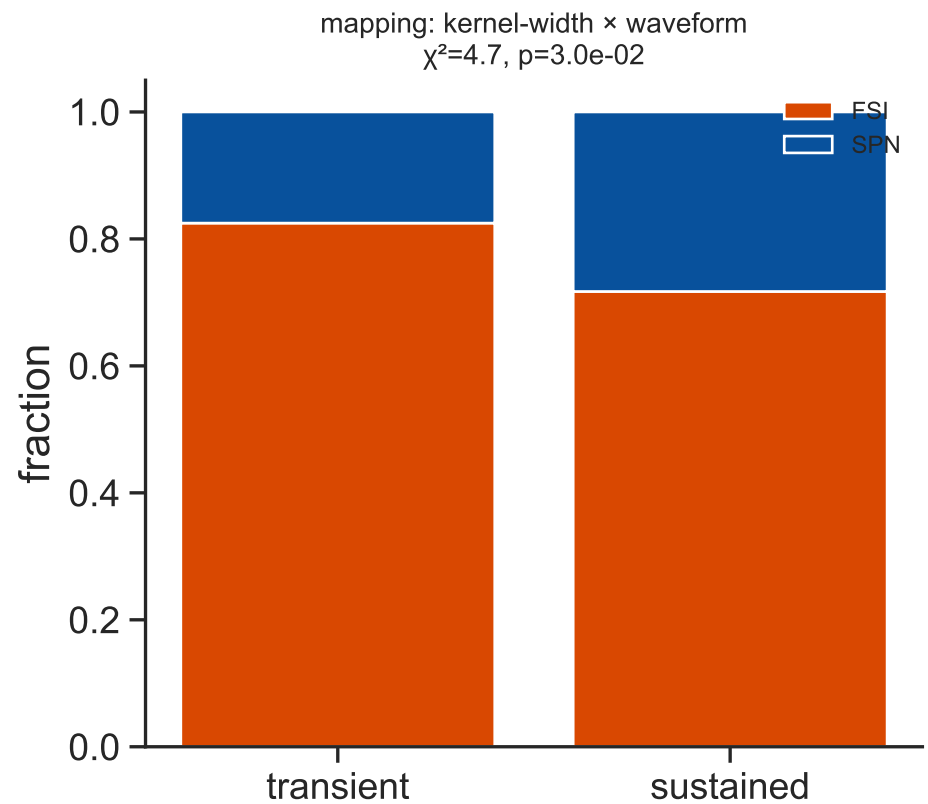
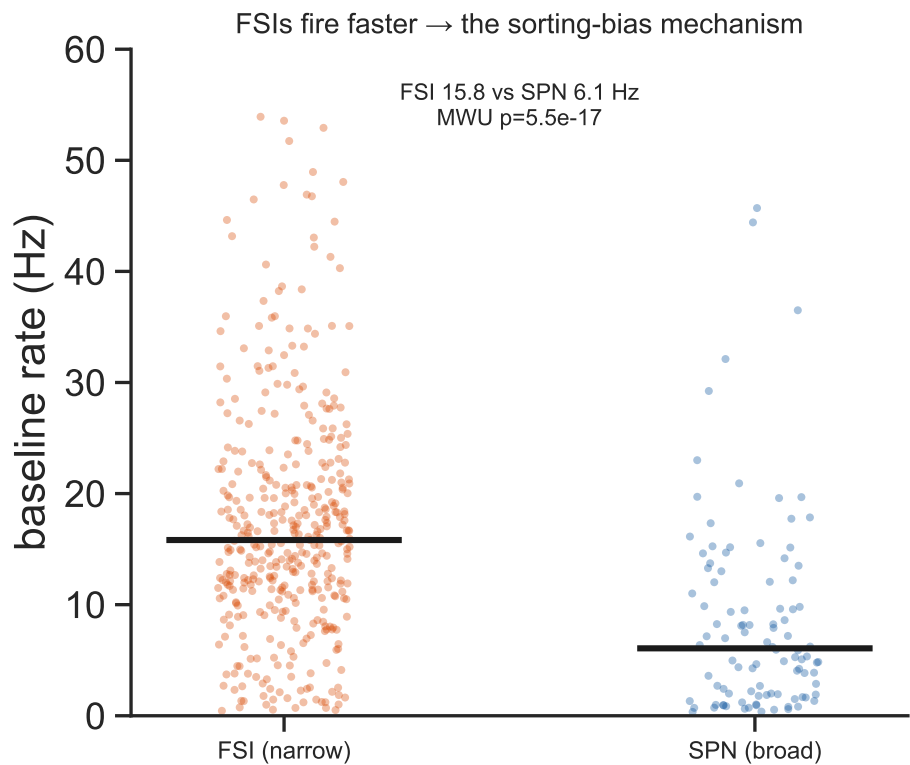
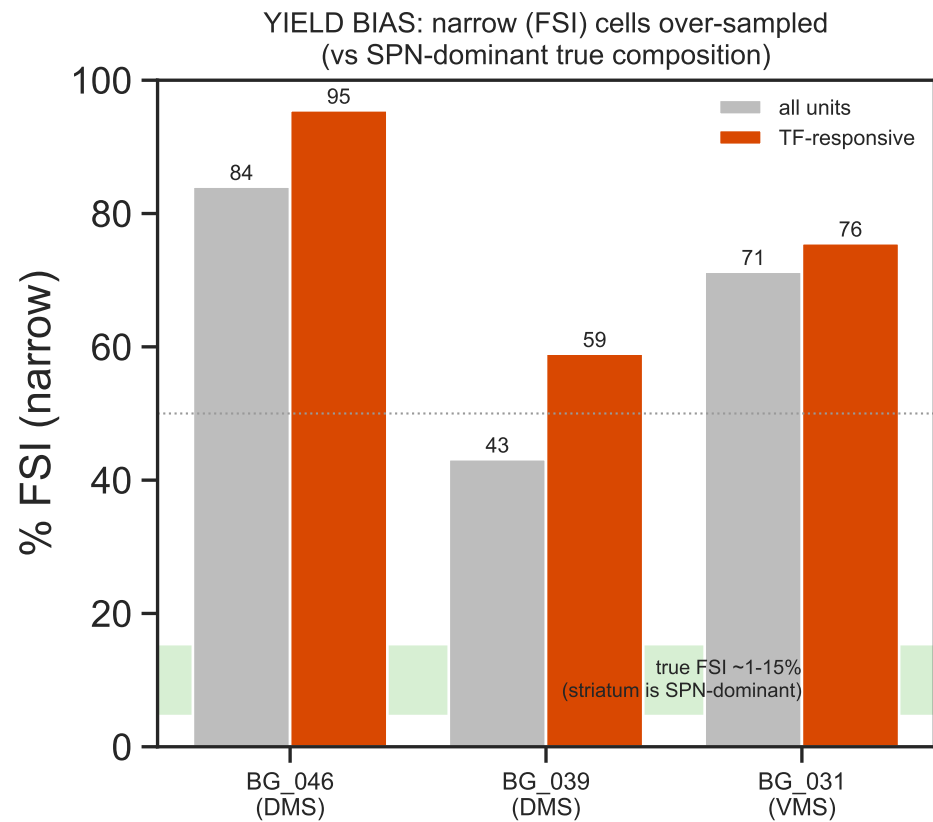


TF-cell kernel width vs FSI/SPN waveform type, and the narrow-cell over-sampling confound
does the transient→sustained outcome-coupling gap survive rate-matching & within-cell-type controls?



```
responsive cells=520; with FSI/SPN label=491 (94% join)
BG_046 all-units: {'FSI': 3769, 'SPN': 716}
BG_039 all-units: {'SPN': 1472, 'FSI': 1117}
BG_031 all-units: {'FSI': 5518, 'SPN': 2224}
[base_hz] FSI med=15.82(n=391) vs SPN med=6.07(n=100) MWU  $p=5.50e-17$ 
[mapping] crosstab class×celltype:
celltype FSI SPN
class
transient 245 52
sustained 71 28
chi2=4.70  $p=3.02e-02$ 
[width-rate] Spearman(base_hz, kernel_fwhm)  $\rho=+0.142$   $p=1.17e-03$ 
[change_on] RAW  $t=0.49/s=1.44$   $p=3.30e-07$  | RATE-MATCHED med- $p=1.12e-04$  ( $\Delta s-t=+0.81\text{Hz}$ , 100% reps  $p<.05$ , n/grp-99)
[hit_ramp] RAW  $t=1.18/s=4.26$   $p=2.09e-07$  | RATE-MATCHED med- $p=1.12e-04$  ( $\Delta s-t=+2.68\text{Hz}$ , 100% reps  $p<.05$ , n/grp-99)
[fa_ramp] RAW  $t=0.81/s=3.85$   $p=4.42e-11$  | RATE-MATCHED med- $p=7.34e-07$  ( $\Delta s-t=+2.78\text{Hz}$ , 100% reps  $p<.05$ , n/grp-99)
[within FSI] change_on:  $t=0.7910364145658269(n245)/s=1.4734356552538408(n71)$   $p=0.0006694487001068222$ 
[within FSI] hit_ramp:  $t=1.614942528735634(n245)/s=4.352617079889811(n71)$   $p=0.0002911227282275486$ 
[within FSI] fa_ramp:  $t=1.0677083333333357(n245)/s=4.070175438596491(n71)$   $p=6.292979792284656e-08$ 
[within SPN] change_on:  $t=0.017317108294930952(n52)/s=1.3511764705882356(n28)$   $p=4.2213883480488813e-07$ 
[within SPN] hit_ramp:  $t=0.2593381446928354(n52)/s=3.9185613890037203(n28)$   $p=6.67171215266705e-06$ 
[within SPN] fa_ramp:  $t=0.255280528052805(n52)/s=3.5599162985352253(n28)$   $p=1.1986231009096956e-05$ 
```